PROJECT PROPOSAL | FOR MANAGEMENT DISCUSSION

Retail Banking Analytics Prototype

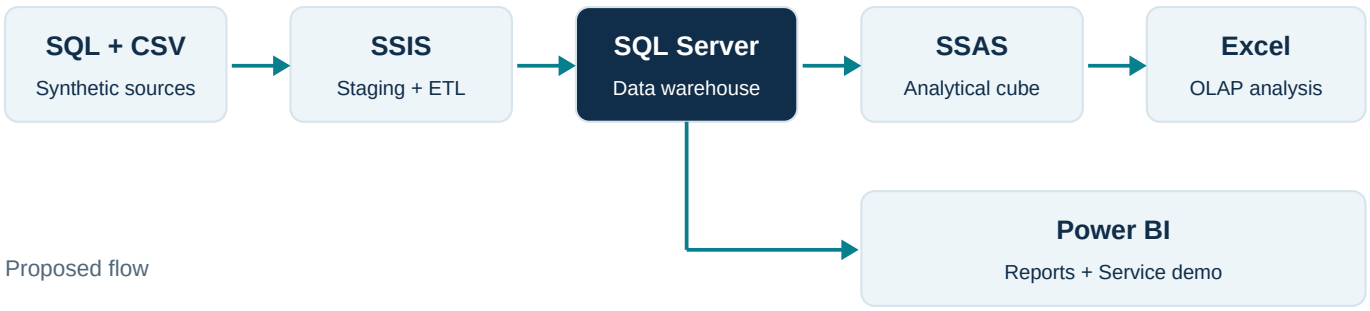
A practical plan for building and explaining a reliable DW/BI system

Build a reproducible demonstration that turns synthetic banking activity into consistent analytical views. Develop programming, database and data-engineering skills through one working system, with a concise report and demonstration at each meaningful milestone.

Current position: Course-material review and initial local Git setup are complete. The data model, pipeline and reports are proposed work.

Proposed initial scope

Customer, Account, Branch and Transaction form the starting scope. Use about one year of meaningful synthetic records and two source types. Final attributes and source allocation will follow the first design exercise.



Proposed flow

Questions the demonstration should answer

How do deposit amounts and transaction counts vary by branch and month? How does activity differ by account type? Which simulated transactions are pending, and how long do completed transactions take?

Tools with a clear purpose

Tool	Responsibility
Python	Generate reproducible synthetic records and independent validation checks.
SQL Server + SSIS	Store the warehouse; extract, transform and load data; handle changes and completion updates.
SSAS + Excel	Create a cube and hierarchy; explore data through OLAP operations.
Power BI + Git/GitHub	Present interactive reports and an online demonstration; keep reviewed version history.

One shared dataset and warehouse support the entire build. Progress follows verified milestones.

DELIVERY AND EVIDENCE

Build in a clear sequence

Complete and validate the warehouse and ETL foundation, then extend it with analytics and reporting.

Stage	Focus	Evidence before proceeding
01	Business scope and source model	Business questions, an ERD, field definitions and explicit assumptions.
02	Reproducible source data	A small Python generator, SQL/file sources and valid relationships across about one year.
03	Warehouse implementation	Declared grain, measures, dimensions, Date and a dimension-history design implemented in SQL Server.
04	SSIS data integration	Verified source-to-staging-to-warehouse flows, lookups, load order and error handling.
05	Completion-update workflow	Separate update input and SSIS package; matched IDs, correct elapsed hours and missing-ID handling.
06	Cube and Excel analysis	A working SSAS cube and hierarchy; roll-up, drill-down, slice, dice and pivot demonstrated in Excel.
07	Reports and handover	Power BI matrix, cascading filters, drill-down and drill-through; Service demonstration and concise report.

Quality checks across the whole system

Reconcile row counts and amounts across sources, staging, warehouse and reports. Validate key relationships, dimension history, rejected rows, completion matching and repeat-run behavior. Check that report filters and drill actions retain the intended context.

What the manager receives

A clear executive summary, architecture and model diagrams, selected screenshots, validation results and a reproducible demonstration. Supporting artifacts include source-generation instructions, mappings, SQL scripts, SSIS packages, cube project, Excel workbook and Power BI reports.

Practical learning rhythm

Learn one concept, build one small piece, verify it, explain the result, then record the change in Git. Expand the system when the current piece is understood.

Project boundary: synthetic demonstration data. Workplace requirements and any production deployment would be defined separately.